



STA 3182 Advanced Distribution Theory

Mini Project Report

Applications of Bi-variate Distribution & Order

Statistics

by

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Task 1

Application of a Bi-variate Distribution

1.1 Full APA citation of the referred paper

George, J., Letha, J., & Jairaj, P. G. (2016). Daily Rainfall Prediction using Generalized Linear Bivariate Model - A Case Study. *Procedia Technology*, 24, 31-38.
doi:10.1016/j.protcy.2016.05.005

1.2 Context of the study

Water resources planning and management are based on precise information about past, present and future precipitation at annually, monthly, and daily time periods. Here, precipitation is referred to any form of water which falls from the atmosphere to the earth's surface. Predicting daily precipitation is a challenging task in water resource management even with the updated climatic models. The first weather generator was introduced by Richardson in order to generate the daily weather sequences. However, that generator failed to explain adequately the length of dry and wet series and had a tendency of underestimating validity of seasonal means or totals. To address those issues different approaches were suggested. As a result, weather generators based on General Linear Models were introduced.

George et al. (2016) developed a multisite weather generator based on atmospheric predictors and historical data using a GLM in the catchment of Idukky reservoir in Kerala, India to simulate the daily rainfall series. The proposed model was used to simulate the daily rainfall series for the missing period from 1981 to 1995 using the atmospheric predictors derived from re-analysis data and to simulate daily rainfall series for a future period of 2013 to 2025 using the predictors derived from the Global Climate Model (GCM) outputs. According to this, the developed model was used as a weather generator to simulate the daily rainfall series for both past and future periods.

1.3 Role of bi-variate distribution

In that study, the bivariate distribution was used to jointly model the precipitation and temperature due to the dependence between these two variables. Furthermore, in that model, it can be seen that one variable was assigned as a covariate for the other variable to link these variables together. This implies that, the bivariate framework allows one variable to influence the conditional distribution of the other variable, by incorporating their correlation. The key properties of the bivariate distribution such as covariance, and conditional dependence were used in that model in order to link the two variables while modelling each marginal distribution appropriately. Hence, the role of the bivariate distribution in that study was to describe the joint behaviour and the interdependence between the rainfall and the temperature, supporting more accurate simulations of daily rainfall patterns.

1.4 Mathematical and conceptual insight

According to the study, daily rainfall is simulated using GLM for occurrence and intensity of rainfall and temperature in three stages. In this study, GLM models a $n \times 1$ random vector Y based on p covariates, whose values are arranged in $n \times p$ matrix X . Each Y depends on p covariates through a linear equation $g(\mu_i) = X_i\beta = \eta_i$. Here, μ is the mean vector which is related to X as in this linear relationship. Here Y_i is the daily rainfall and it is assumed to be generated in the form of the above stated equation because the distribution of Y has the mean vector μ . Furthermore, $g(\mu_i)$ is the link function. Each Y_i is assumed to have a distribution from the exponential family. According to the study,

A logistic regression model is used to predict the probability of a rainfall occurrence. The corresponding equation is given as;

$$\ln\left(\frac{p_i}{1-p_i}\right) = X_i\beta$$

Here, p_i denotes the probability of rainfall on i^{th} day, X_i denotes the corresponding predictor vector of atmospheric circulation patterns and β denotes the coefficient vector.

To predict the amount of rainfall on a rainy day, the gamma distribution is used. The corresponding equation to predict the mean rainfall on a rainy day is given by;

$$\ln(\mu_i) = Z_i \alpha$$

Here, μ_i is the intensity of rainfall at a site on i^{th} day and Z_i is the predictor vector based on atmospheric circulation patterns.

For the parameter estimation, Maximum Likelihood Estimation (MLE) method was used. Mean sea level pressure, u-wind components, v-wind components, air temperature, and relative humidity were used as the atmospheric predictors in this model. Moreover, the model used various covariates to account for the effects of site, year, month, and day. Models were calibrated from 1995 to 2013 and validated from 1981 to 1995 with 10 simulations. Residual checks, Q-Q plots for residuals, and comparisons of simulated and observed data were applied as the diagnostic tests.

1.5 Assumptions

The study assumed that the amount of rainfall on a rainy day can predict by a gamma distribution but this may not be accurate on the heaviest rainy days. It also assumed that a normal heteroscedastic model was appropriate for daily temperature simulation.

1.6 Limitations

One of the limitations is that, this model did not fully capture 20% of the peak values of historical rainfall due to the inconsistency in the data set. Another limitation is, even though this study was developed as a multisite analysis, data from a single station was considered due

to missing records at the other sites. Therefore, this model may be weak or not suitable to apply in multisite.

1.7 Extensions that could improve the method

This model can be improved by including more stations around the catchment. And also, by omitting the stations with high residual error, the model can be redefined. This model's performance can be improved by adding more covariates which have significant impacts on rainfall.

Task 2 (A)

Application of Order Statistics

2.1.1 Full APA citation of the referred paper

Busababodhin, P., Chiangpradit, M., Papukdee, N., Ruechairam, J., Ruanthaisong, K., & Guayjarernpanishk, P. (2021). Extreme Value Modeling of Daily Maximum Temperature with the r-Largest Order Statistics. *The Journal of Applied Science*, 20(1), 28-38. doi:10.14416/j.appsci.2021.01.003

2.1.2 Context of the study

The study addressed the problem of modelling the daily maximum temperature in the first three highest temperature provinces namely Mae Hong Son, Nan, and Uttaradit in north of Thailand. In this study, data from 1948 to 2018 was used. Furthermore, this study identified the gap that many researchers analysed only the single largest value in other words, first order statistic per year. To address that gap, in this research, it was applied r-largest order statistics applying with of daily maximum temperature of northeast Thailand to study extreme value modelling. Moreover, the ultimate goal of this study was to provide highest benefits for dealing with the temperature problems by achieving more accurate results and to provide a valuable tool for climatic risk management and disaster management.

2.1.3 Role of order statistics

This study employed the r-largest order statistics to create a model with the Generalized Extreme Value (GEV) distribution with different orders like $r = 2, 3, \dots, n$. Here the number of data was denoted by n . Those values were selected for each meteorological station based on a deviance statistic test. Then, after comparing the models using those deviance statistics, they identified the most suitable r value for each meteorological station.

2.1.4 Mathematical and conceptual insight

The methodology of this study was based on the extreme value distribution theory, more specifically the GEV distribution. The GEV distribution was defined by three parameters namely location parameter (μ), scale parameter (σ), and shape parameter (ξ). The sign of the shape parameter was used to determine the type of the distribution. If the value of ξ approaches zero, then the GEV is called as the Gumbel distribution. If the value of ξ is greater than zero, then the GEV is called as the Fréchet distribution, and if the value of ξ is less than zero, then the GEV is called as the Weibull distribution. In this study, this concept was extended to the r -largest order statistics. In order to find out the most appropriate model, as the first step they cleaned and analyzed the daily temperature data from seven meteorological stations in the Mae Hong Son, Nan, and Uttaradit provinces in north Thailand. To model the data, they used R programming language with Extreme Value Analysis (EVA) package. Then, by comparing the models using the deviance statistics test, the most appropriate value for r was selected. Moreover, to estimate the model parameters μ , σ , and ξ , the Maximum Likelihood Estimation (MLE) method was applied. After that, the most suitable distribution out of the three distributions; Gumbel, Fréchet, and Weibull was identified based on the value of the shape parameter ξ . Finally, to predict the future extreme temperatures, the return levels for different return periods such as 2, 5, 10, 20, and 50 years were calculated.

2.1.5 Assumptions

In this study, the data were assumed to be independent and identically distributed. Also, the GEV- r model assumed that the upper tail of the temperature distribution followed an extreme distribution. Another assumption made in this study is, the stationarity of the data without considering the structural trend over the past 35 years. Furthermore, in this study, it was assumed that MLE method was the best method to estimate the model parameters.

2.1.6 Limitations

One of the limitations in this study is, the independent and identically distributed assumption of the data. This i.i.d assumption may not valid for long-term climatic data due to the trends in the climatic changes. Furthermore, the study considered only three provinces in north Thailand. This can lead the results and insights made from this study to be inappropriate or misleading due to different geographical variations in other provinces. And also, the data were assumed to be stationary but this assumption may not be valid for the long-term climatic data due to the trends in climatic changes. Another limitation of the study is, the method used in this study focused only on marginal distributions of extreme temperatures without considering the correlations among the meteorological stations.

2.1.7 Extensions that could improve the method

To improve the model, it can be used non-stationary models by allowing GEV parameters to vary with the time and covariates such as humidity and global warming indicators. To improve the accuracy and the validity of the model, this study can be extended to the other provinces as well because in this study, it was mentioned that provinces like Lampang, Tak, Sukhothai, Nakhon Sawan, and Kanchanaburi which were not considered in this particular study had increasing temperatures.

Task 2 (B)

Novel Application of Order Statistics

Title

Preference modelling of grocery items in supermarkets using order statistics.

Context

Minimizing the time spent for a grocery shopping trip by a shopper is a real-world problem because a shopper often has a long list of grocery items, and the stores or the supermarkets are often not organized efficiently. As an example, milk is stored at the back of the store. Due to this reason, sometimes customers have to spend more time during their shopping trip. This is problematic, for both type of customers who have long lists of grocery items and who need to buy only few items.

Order statistics can be used to optimize the grocery shopping time by learning from past trips. This can be used to re-order the items in a supermarket or a store by identifying the highest priority items and so on.

Reason of using of order statistics in this application

Order statistics provide an appropriate and simple framework for the above mentioned problem. The maximum and minimum order statistics of past purchases can be used to identify the most and least frequently bought items. This helps in making a new list that the grocery items have to be stored effectively and efficiently by prioritizing the customers' needs. Furthermore, to address the above mentioned problem, order statistics is suitable because of the simplicity of the computations. Because for a short list, computing the minimum, maximum, and median is not that much difficult. These computations can be done quickly with the relevant software. There are long lists as well as short lists and lists which are in between these two lists. So, a

single purchase of an unusual grocery item can be act as an outlier and this can affect the prioritization of the new ordered list. But when using order statistics such as median, this effect can be eliminated. Moreover, the purchasing habits of customers are not neat and these data are not normally distributed. However, this issue can be addressed by using order statistics because order statistics themselves can be defined for any sample regardless of underlying distribution.

Proposed methodology

A smart home app or a grocery store's app can be introduced to the customers to obtain the past purchased data, either from uploading the receipts or a linked store account. To store these past data, a simple database is needed. After that, for each grocery item, the system calculates its purchase frequency. Purchase frequency can be calculated by dividing the number of times that particular item was purchased by the total number of grocery shopping trips. And also, if a customer creates a new grocery list, then the system looks at how often that customer has bought each item before and arranges the items from most frequently purchased to least frequently purchased. Furthermore, the system can use the median purchase frequency as a cutoff. Then the grocery items with purchase frequencies above the median are prioritized at the top of the list, while less frequent items are moved to the bottom of the list.

Feasibility

This approach is simple to implement. The functionality requires a database of past purchases and a basic sorting algorithm which works well within the capabilities of any smartphone.

Impacts

Customers can reduce the grocery shopping time with a more effective and efficient path when the grocery items are re-ordered according to the priority of the customers. Also, this application is personalized to the user's habits. Then it is useful in providing a more useful tool than an unsorted list.